

Simulation-Based Inference

Simulation-based tests

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Frequentist perspective on probability

Probability as the limit of relative frequencies

Examples:

- In 92% of depressive patients, we find that psychotherapy improves their mood.
- In 80% of the experiment's repetitions, we find the drug to have a significant effect.
- In 30% of the days, it will rain in Dortmund.

Question: What is the difference between the last example and the following statement?

- Tomorrow it will rain with probability 30%.

Example: Intelligence of students

Research question: Are students on average more intelligent than people from the general population?

Information about the general population:

- The IQ values are scaled such that the mean is $\mu_G = 100$ and standard deviation is $\sigma_G = 15$
- The IQ values are normally distributed

Information about the student population:

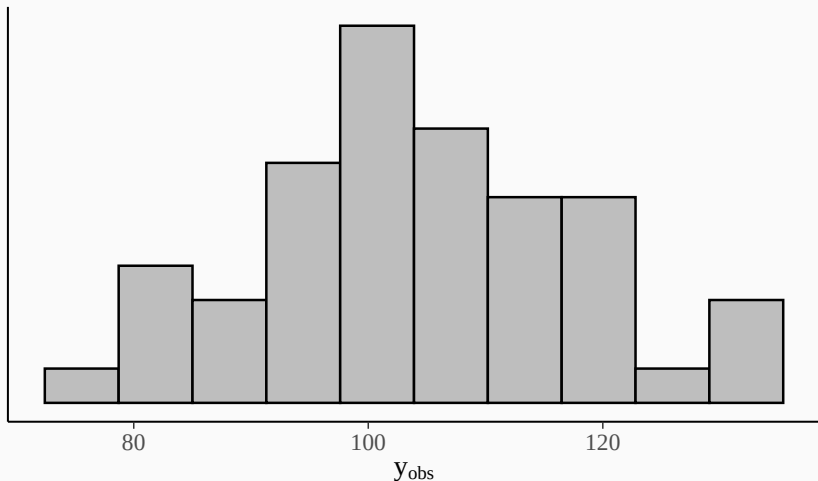
- The mean value μ_S is unknown (research question)
- The standard deviation is known with $\sigma_S = 15$ (simplification)
- The IQ values are normally distributed (simplification)

We derive from the research question

$$H_0 : \mu_S \leq \mu_G \quad \text{vs.} \quad H_1 : \mu_S > \mu_G$$

- μ_S is the (unknown) expected value of the students
- $\mu_G = 100$ is the expected value in the general population
- H_1 is the substantive hypothesis
- H_0 is the logical counter-hypothesis
- We test by rejecting H_0 (null hypothesis significance test; NHST)

Dataset: 50 randomly selected students



Empirical mean $\bar{y}_{\text{obs}} = 104.5$ as an estimate of μ_S

The p -value

Definition (semi-formal): The p -value is the probability under H_0 to obtain the observed value of the test statistic or a more extreme value in the direction of H_1 .

Key aspects of the definition:

- The p -value assumes that the H_0 is true
- The p -value refers to a specific test statistic
- The p -value compares the observed value of the test statistic with its theoretical distribution under H_0
- The definition of *extreme* values depends on the H_1

For our example:

- The test statistic is the mean value $\bar{y}$
- More extreme values are *greater* values

The generative model under H_0

Under H_0 , the $N = 50$ IQ values of the students are

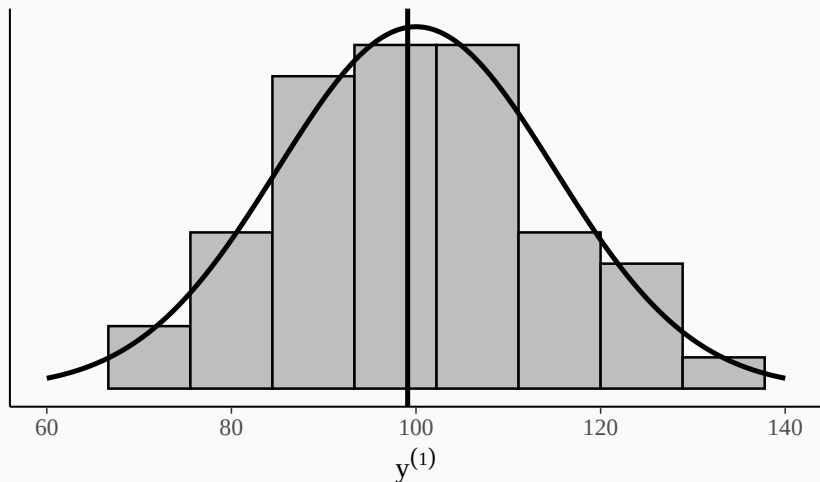
- independent of each other
- normally distributed
- with $\mu_G = 100$ and $\sigma_G = 15$

Mathematical notation:

$$y_i \sim \text{normal}(100, 15) \quad \text{for all } i = 1, \dots, N$$

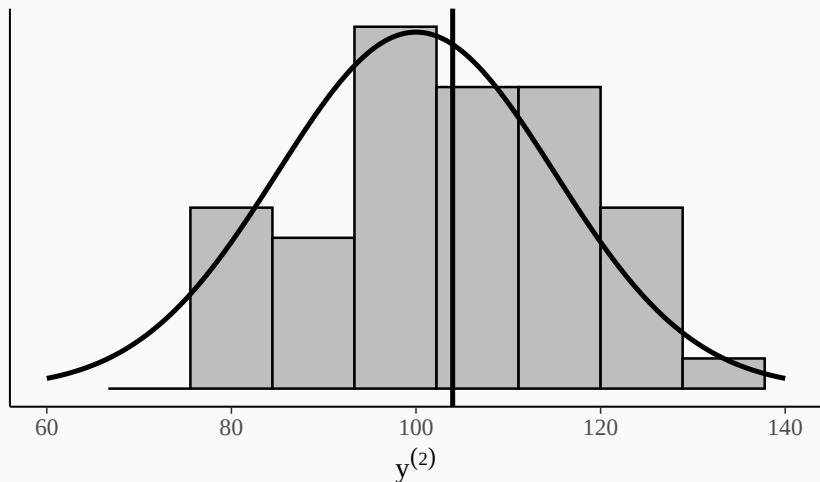
We therefore have a *generative model* of H_0 from which we can simulate!

First simulated dataset from H_0 with $N = 50$



The mean of the first simulated dataset is $\bar{y}^{(1)} = 99.1$

Second simulated dataset from H_0 with $N = 50$



The mean of the second simulated dataset is $\bar{y}^{(2)} = 104$

Repeatedly simulating datasets

We repeat the simulations of datasets S times

S should be very large if possible, say $S = 10000$

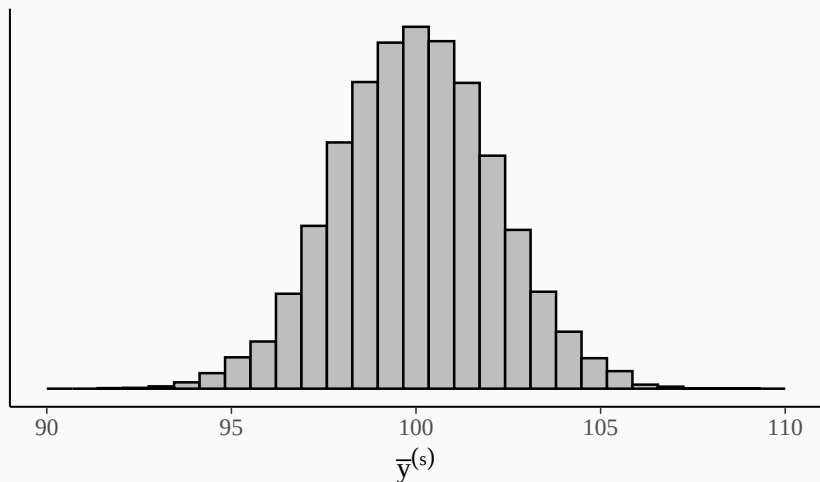
For every $s = 1, \dots, S$:

- Draw values $y_i^{(s)} \sim \text{normal}(100, 15)$ for $i = 1, \dots, N$
- Compute the mean $\bar{y}^{(s)}$ of the N values $(y_1^{(s)}, \dots, y_N^{(s)})$

Result: A set of S simulated means

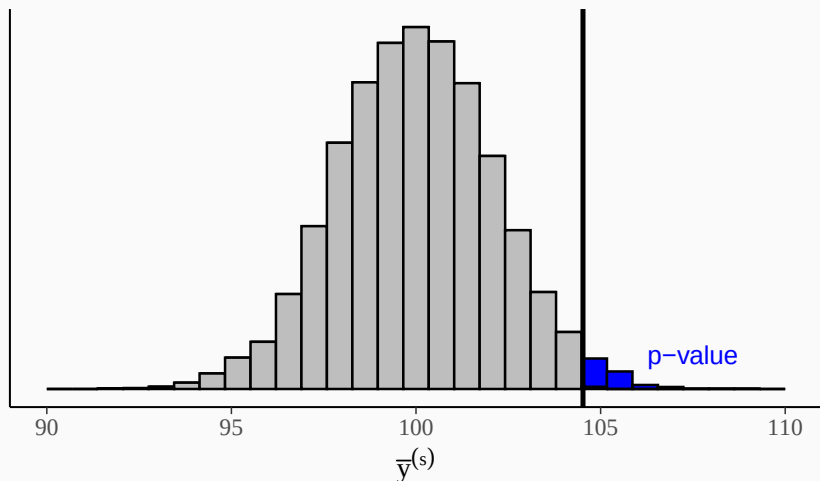
$$\{\bar{y}^{(1)}, \dots, \bar{y}^{(S)}\} =: \{\bar{y}^{(s)}\}_{s=1}^S =: \{\bar{y}^{(s)}\}$$

Sampling distribution of the mean under H_0



Histogram of the means based on $S = 10000$ simulated datasets

Comparison with the empirical mean



The p -value evaluates to $p = P(\{\bar{y}^{(s)}\} \geq \bar{y}) \approx 0.02$

Definition: A statistical model from which we can simulate data is called a (frequentist) *generative model*.

We write

$$y \sim M = M(\theta, \psi)$$

- θ : for parameters of interest
- ψ : nuisance parameters
- y : data

We will denote the generative model under H_0 as M_0

Question: Do you have an example of a statistical model that is not generative?

The sampling distribution

Definition: Let $T = T(y)$ be a function of data y , then we call T a *test statistic*.

Definition: If $y \sim M$, the distribution $p(T(y) \mid \theta, \psi)$ is called *sampling distribution* of T under generative model M .

Example: If $M = \text{normal}(\theta, \psi)$, then the mean $T(y) = \bar{y}$ of N draws y_i from M has an analytic sampling distribution:

$$\bar{y} \sim p(\bar{y} \mid \theta, \psi) = \text{normal}(\theta, \psi/\sqrt{N})$$

Simulation-based version: For each simulated dataset $y^{(s)}$, compute $T(y^{(s)})$. Then plot the histogram over the resulting set $\{T(y^{(s)})\}_{s=1}^S$.

The p-value (now more formal)

Definition: Let $T(y)$ be a test statistic and $H_0 : \theta \leq \theta^*$ for some fixed value θ^* without loss of generality. The probability $P(T(y) \geq T(y_{\text{obs}}) \mid \theta = \theta^*, \psi)$ for data $y \sim M_0$ is called *p-value* of $T(y_{\text{obs}})$ given H_0 .

Opposite-sided or two-sided p-values can be defined analogously.

Simulation-based version: The simulation-based *p-value* is the (empirical) percentage of values in $\{T(y^{(s)})\}_{s=1}^S$ for which $T(y^{(s)}) \geq T(y_{\text{obs}})$:

$$p = \frac{1}{S} \sum_{s=1}^S \mathbb{I}(T(y^{(s)}) \geq T(y_{\text{obs}})),$$

where $\mathbb{I}$ is the indicator function.

Example: Simulation-based tests of the standard deviation

Consider the example of estimating the standard deviation σ , which is a measure of variability in the data.

The empirical standard deviation is usually defined as:

$$\hat{\sigma}(y) = \sqrt{\frac{1}{(N-1)} \sum_{i=1}^N (y_i - \bar{y})^2}$$

If the data are normally distributed, $\hat{\sigma}(y)$ is consistent and asymptotically unbiased estimator of the true SD σ .

Question: Why is this estimator only *asymptotically* unbiased?

Note: Under normality, the Variance estimator $\hat{\sigma}^2$ (square of $\hat{\sigma}(y)$) follows a scaled χ^2 distribution.

Research question: Is the SD students' IQ scores smaller than 15?

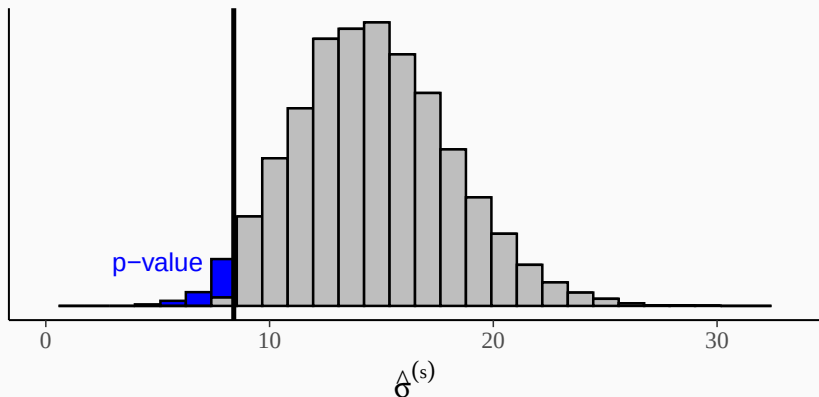
$$H_0 : \sigma \geq 15 \quad \text{vs.} \quad H_1 : \sigma < 15$$

We use only a sample of $N = 10$ students but otherwise assume the same overall scenario as previously.

The empirical SD of our small sample evaluates to $\hat{\sigma}(y_{\text{obs}}) = 8.4$.

Sampling distribution of the standard deviation

- Draw values $y_i^{(s)} \sim \text{normal}(100, 15)$ for $i = 1, \dots, N$
- Compute the SD $\hat{\sigma}^{(s)}$ of the N values $(y_1^{(s)}, \dots, y_N^{(s)})$



The p -value evaluates to $p = P(\{\hat{\sigma}^{(s)}\} \leq \hat{\sigma}(y_{\text{obs}})) \approx 0.03$.

Example: Simulation-based tests of the skewness

The sampling distributions of mean and variance of normally distributed variables are analytically known. So for these cases, we won't really benefit from a simulation-based test.

Let's consider the more complicated example of estimating the skewness γ , which indicates whether on tail of the distribution is fatter than the other.

A common test statistic for estimating the skewness is given by:

$$\hat{\gamma}(y) = \frac{\sum_{i=1}^N (y_i - \bar{y})^3 / N}{\left(\sum_{i=1}^N (y_i - \bar{y})^2 / N \right)^{3/2}}$$

If the data are normally distributed, this $\hat{\gamma}(y)$ is known to be an unbiased and consistent estimator of the true skewness $\gamma = 0$.

For finite samples, the sampling distribution of $\hat{\gamma}(y)$ is non-analytic.

Intelligence of students: skewness

Research question: Is the distributions of students' IQ scores symmetric or skewed?

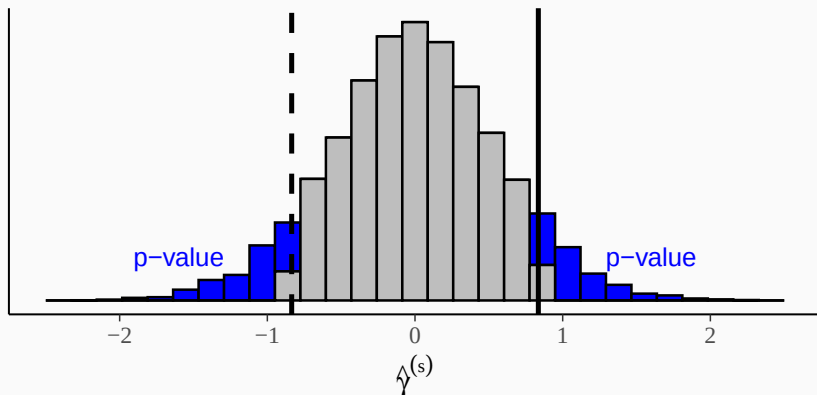
$$H_0 : \gamma = 0 \quad \text{vs.} \quad H_1 : \gamma \neq 0$$

We use only a sample of $N = 10$ students but otherwise assume the same overall scenario as previously.

The empirical skewness of our small sample evaluates to $\hat{\gamma}(y_{\text{obs}}) = 0.84$.

Sampling distribution of the skewness

- Draw values $y_i^{(s)} \sim \text{normal}(100, 15)$ for $i = 1, \dots, N$
- Compute the skewness $\hat{\gamma}^{(s)}$ of the N values $(y_1^{(s)}, \dots, y_N^{(s)})$



The p -value evaluates to $p = P(\{|\hat{\gamma}^{(s)}|\} \geq |\hat{\gamma}(y_{\text{obs}})|) \approx 0.15$.

Unknown nuisance parameters

Sampling from $M_0(\theta, \psi)$ requires all parameters to be specified, that is, both the parameters of interest θ and the nuisance parameters ψ .

While θ is specified by H_0 , ψ is not. Ideally, we would like to design a test that works *independently* of the value of ψ .

Question: Any ideas how to achieve that?

Example continued: Intelligence of students

Consider again our initial example. Now assume that the standard deviations σ_G and σ_S are unknown and may also be unequal.

The sampling distribution of the test statistic $\bar{y}$ depends on σ_G .

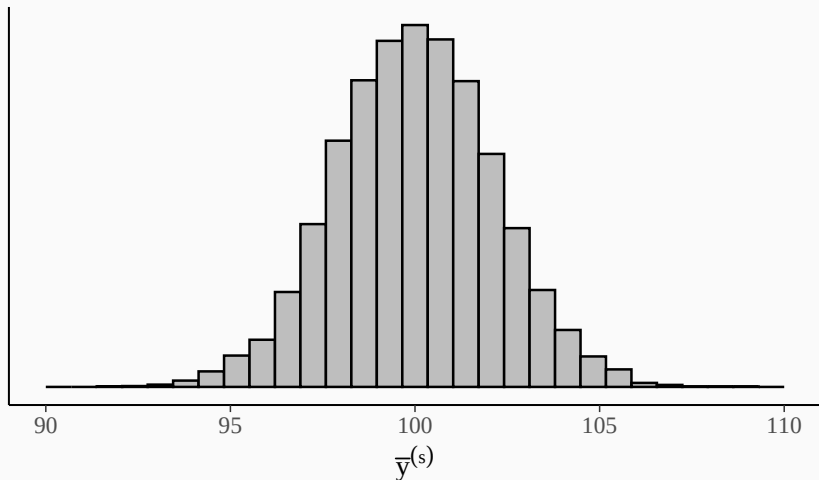
The sampling distribution of the test statistic

$$Z(y) = \frac{\bar{y} - \mu_G}{\hat{\sigma}},$$

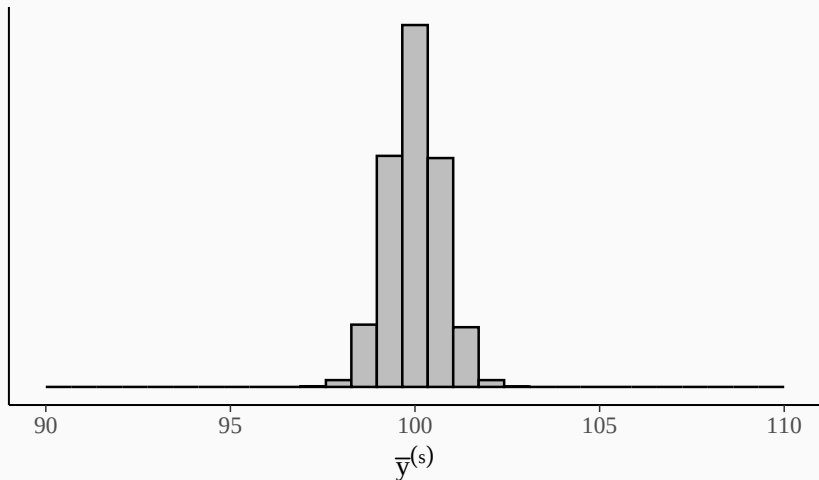
is independent of σ_G .

Here, $\mu_G = 100$ is the IQ mean in the general population and $\hat{\sigma} = \hat{\sigma}(y)$ is the empirical standard deviation of data y .

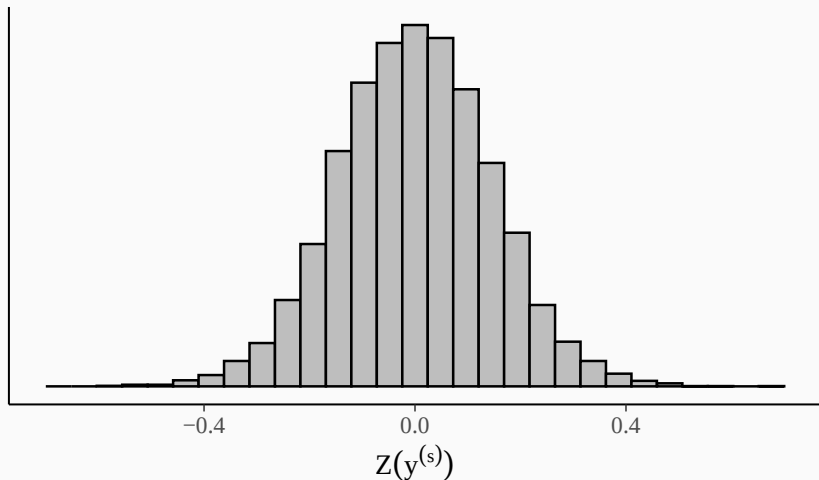
Sampling distribution of $\bar{y}$ under $M_0(100, 15)$



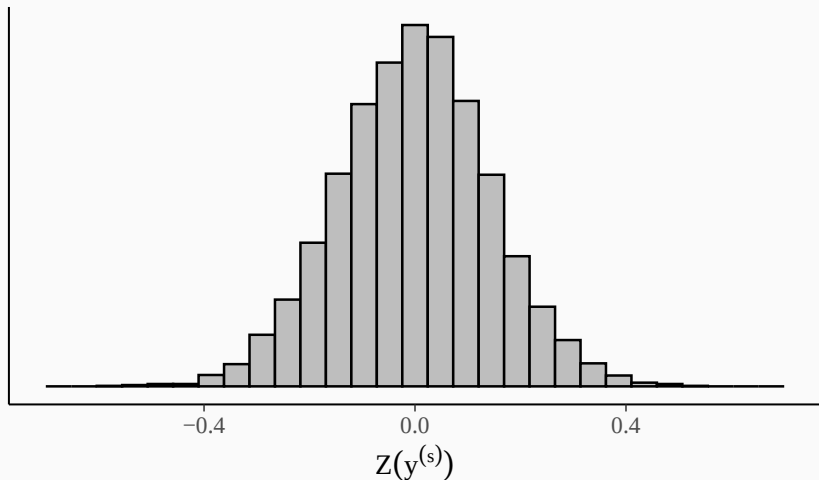
Sampling distribution of $\bar{y}$ under $M_0(100, 5)$



Sampling distribution of Z under $M_0(100, 15)$



Sampling distribution of Z under $M_0(100, 5)$



Confidence intervals (CIs)

Definition: Let θ^* be the true value of the parameters of interest and α the nominal Type-I error rate. An interval of the form $[\theta_l(y), \theta_u(y)]$ is a (calibrated) *confidence interval* for θ^* , if $\theta^* \in [\theta_l(y), \theta_u(y)]$ with probability $1 - \alpha$ for data $y \sim M_0$.

Question: How can we define simulation-based confidence intervals?

Intervals over the sampling distribution

Two-sided intervals: Given α , compute quantiles $Q_{\alpha/2}$ and $Q_{1-\alpha/2}$ of the sampling distribution $p(T(y) \mid \theta^*, \psi)$.

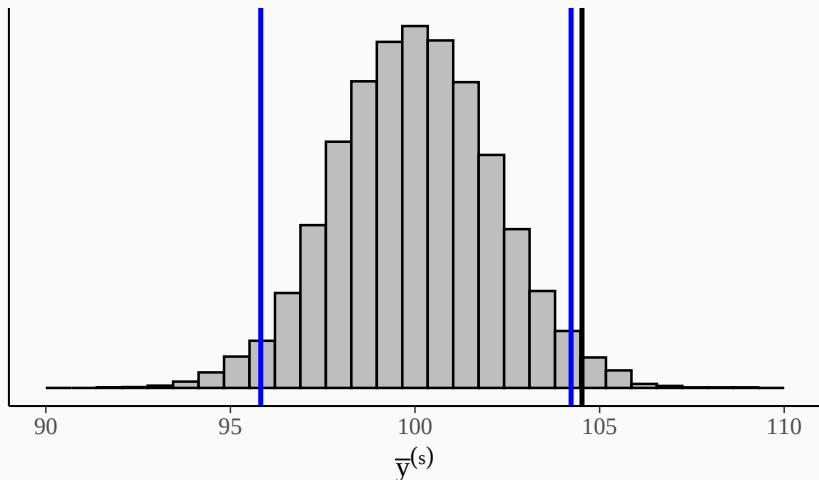
Then we know that the interval $[Q_{\alpha/2}, Q_{1-\alpha/2}]$ contains $T(y)$ with probability $1 - \alpha$ for $y \sim M_0$.

Checking if $T(y_{\text{obs}})$ is within vs. outside of $[Q_{\alpha/2}, Q_{1-\alpha/2}]$ is equivalent to checking if $p > \alpha$ vs. $p \leq \alpha$ for a two-sided p -value.

One-sided intervals: Use $[Q_0, Q_{1-\alpha}]$ or $[Q_\alpha, Q_1]$

Simulation-based version: Compute the empirical quantiles from the set of simulated test statistics $\{T(y^{(s)})\}_{s=1}^S$

IQ example continued: Simulation-based interval



Black: Empirical mean $\bar{y}_{\text{obs}}$, blue: 95% simulation-based interval.

Question: Do sampling distribution intervals help us define simulation-based confidence intervals?

For more complex research questions, we have to actually *fit* statistical models to perform hypothesis testing, instead of relying on closed-form expression for hypothesis tests.

Example: The logistic regression model

$$y_i \sim \text{Bernoulli}(\pi_i) \quad \pi_i = \text{logistic}\left(\sum_{k=0}^K b_k x_{ik}\right)$$

has no closed form estimators of the coefficients b_i . Standard (iterative) estimators, such as MLE, have a sampling distribution that is normal only asymptotically (for large enough N).

Example: Predicting heart disease with logistic regression

Predicting the probability of having a heart disease by age and sex. For illustration, we use only a small data subset of $N = 20$ observations.

First few rows of the data:

disease	zage	sex
1	0.7136023	male
0	-1.4964070	female
1	1.7081065	male
0	-1.1649056	male
0	0.4926014	male
1	0.9346033	female

Fitting a logistic regression model

We fit the following model M_1 to the small heart disease data:

$$\text{disease}_i \sim \text{Bernoulli}(\pi_i) \quad \pi_i = \text{logistic}(b_0 + b_1 z_{\text{age},i} + b_2 \text{sex})$$

```
fit_obs <- glm(disease ~ zage + sex, data=heart_small, family=binomial())
summary(fit_obs)
```

Call:

```
glm(formula = disease ~ zage + sex, family = binomial(), data = heart_small)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-0.6672	1.0709	-0.623	0.5332
zage	1.0459	0.5978	1.749	0.0802 .
sexmale	1.1226	1.2269	0.915	0.3602

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Simulating from a logistic regression model

We will test the the following hypothesis:

$$H_0 : b_1 = 0 \quad \text{vs.} \quad H_1 : b_1 \neq 0$$

The generative model M_0 corresponding to H_0 is

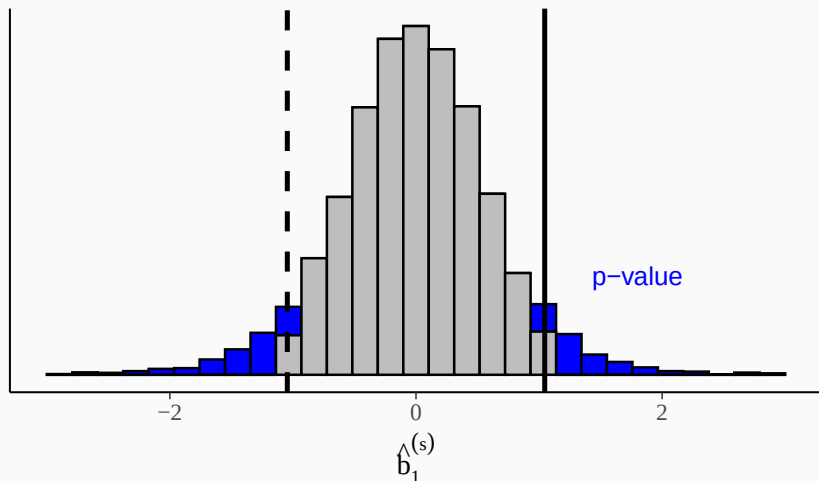
$$\text{disease}_i \sim \text{Bernoulli}(\pi_i) \quad \pi_i = \text{logistic}(b_0 + b_2 \text{sex})$$

With nuisance parameters b_0 and b_2 unspecified. We choose to fix them to the estimates provided by the real data model: $\hat{b}_0 \approx -0.67$ and $\hat{b}_2 \approx 1.12$.

We then fit the full model M_1 to the simulated responses with the predictor values taken from the real data.

Question: Why use the predictor values from the real data instead of also simulating them?

Sampling distribution of the age regression coefficient



The p -value evaluates to $p = P(\{|\hat{b}_1^{(s)}|\} \geq |\hat{b}_1(y_{\text{obs}})|) \approx 0.11$.

Summary

If the H_0 implies a generative model M_0 , we can obtain a *well calibrated* test for *any* data-based quantity T : We repeatedly simulate data $y^{(s)}$ from M_0 and compare the set of simulated results $\{T(y^{(s)})\}_{s=1}^S$ with the empirical value $T(y_{\text{obs}})$.

Limitations of simulation-based tests:

- (1) Simulations require the *full* specification of M_0 . However, sampling distributions often depend on nuisance parameters, which are not given by the hypotheses.
- (2) Simulation-based confidence intervals are hard to define: We have to fix the true parameter value θ^* to what is *assumed* under H_0 . But for confidence intervals, we would have to know what parameter values are plausible given the *observed* data y_{obs} .
- (3) Model-based test statistics can be computationally expensive: Fitting an already *slow* model S times will be *very slow* if S is large.